ACC250: Intro to Financial Accounting

Ch11. Shareholders' Equity

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Learning Objectives

- Record **stock issuance** as **common stock** (contributed capital).
- Distinguish **authorized**, **issued**, **treasury**, and **outstanding** shares.
- Account for **cash dividends** and **treasury stock**.
- Compute **shareholders' equity**: contributed capital − treasury stock + RE.
- Compute **ROE**, **EPS**, and the **P/E ratio**.

Table of Contents

1. Corporations
2. Shares
3. Dividends
4. Shareholders' Equity
   - Contributed Capital
   - Treasury Stock
   - Retained Earnings
5. Ratio analysis
   - Return on Equity (ROE)
   - Earnings Per Share (EPS)
   - Price-to-Earnings (PER) Ratio
6. Concluding illustration
7. Appendix: Par Value and APIC (Not for exam)



Corporations (Recall: Ch1)

Corporations

a business structure that separates ownership from management and provides limited liability to its owners.

- **Separate legal entity**: the corporation is distinct from its owners (shareholders).
- **Limited liability**: owners can lose at most what they invested.

EX. You invested \$10K in a corporation that goes bankrupt owing \$1M. How much can you lose? _____

Corporations (Recall: Ch1)

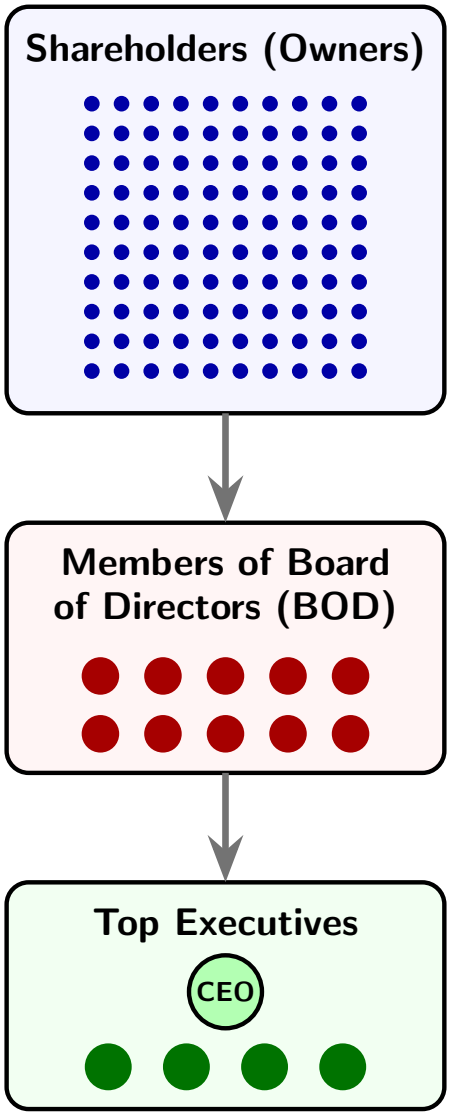
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Corporate Structure



Vote only on major corporate matters (e.g., election of the BOD; M&A).

Involved in important decision-making (incl. CEO appointment/firing).

Responsible for the day-to-day operations.

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Shares[2]

- Corporations issue **shares** to shareholders and receive _____ in return.
- **Shares (a.k.a. stocks)** represent _____; a claim on the corporation's **net assets**[1] and **earnings**.
- If you own 1% of **the shares**, you own _____ of **the corporation**.
- Shares can be bought and sold, allowing investors to **trade ownership**.

[1] Net assets = Assets - Liabilities.

[2] There are more than one type of shares/stocks. We focus on **Common Stock** in this course. Other types include Preferred Stock.

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Shares - Common stocks

Stockholders have:

1. **Voting rights** on important matters (e.g., election of BOD members; M&A).
2. **Right to receive dividends**, if any.[3]
3. **Claims to net assets** in case of liquidation (a.k.a. **residual claims**).

[3] Dividends are **not guaranteed**: the board of directors decides whether to declare them.

EX. A corporation has 1K shares of common stock outstanding and you own 100 shares. The corporation has \$1M in assets and \$800K in liabilities. In case of liquidation,

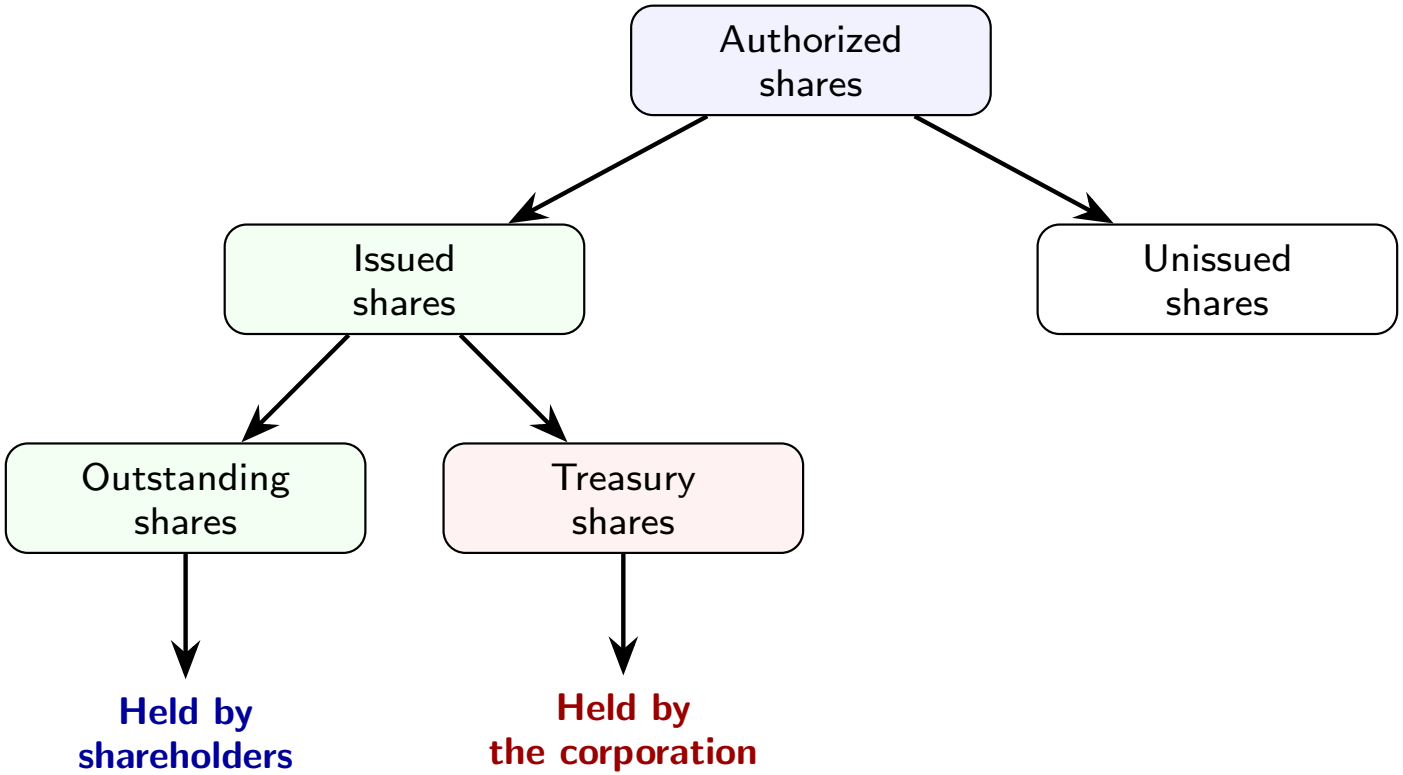
- \$800K would be used to pay off the liabilities.
- The remaining \$200K (residual claims) would be distributed to the shareholders.
- You would be entitled to:

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- \$800K would be used to pay off the liabilities.
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- You would be entitled to:

$$10\% \times (\$1M - \$800K) = 10\% \times \$200K = \$20K$$

Shares - Overview



Shares - Authorized

- Shares must be **authorized** for issuance by the **BOD** before they are actually issued.

EX. The BOD authorized 1M shares for issuance. How much cash is coming in?

— authorization alone is **not** a sale of shares.

Authorized
1,000K

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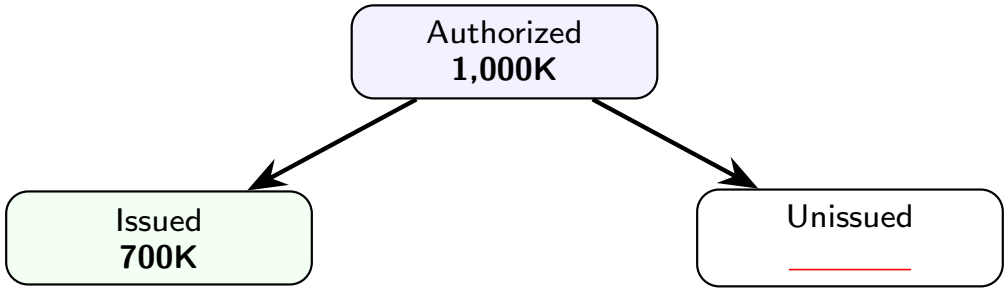
$0 — authorization alone is **not** a sale of shares.

Authorized
1,000K

Shares - Issued

- **Issued shares**: The number of shares that have been **sold to investors**.
- **Unissued shares**: Authorized shares **minus** Issued shares.

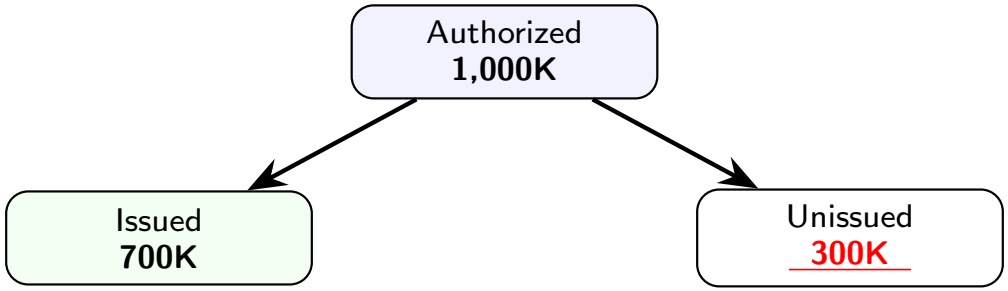
EX. A corporation is authorized to issue 1M shares and has sold 700K shares to investors.



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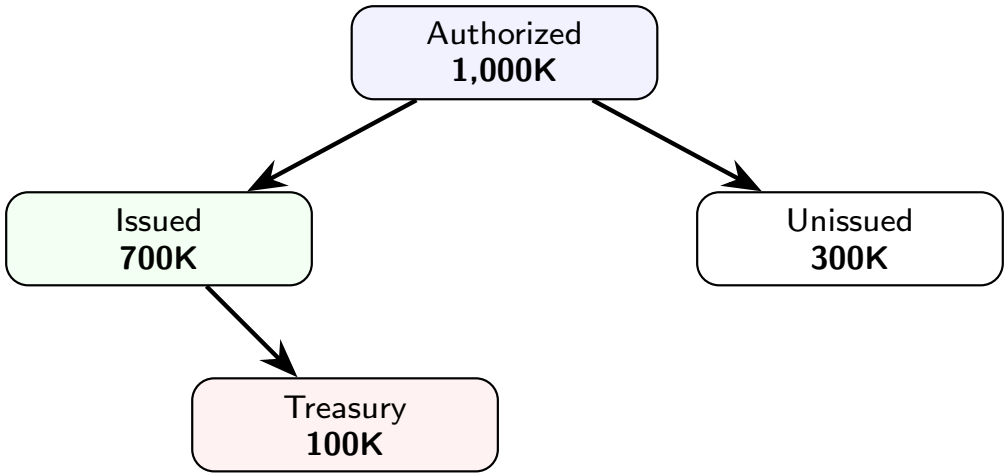
EX. A corporation is authorized to issue 1M shares and has sold 700K shares to investors.



Shares - Repurchased (= Treasury Stock)

- Firms can **repurchase** their own stocks from shareholders.
- It reduces the shares that are **available for trading** in the market.
- The repurchased stocks are **held by the company**: **Treasury Shares**.

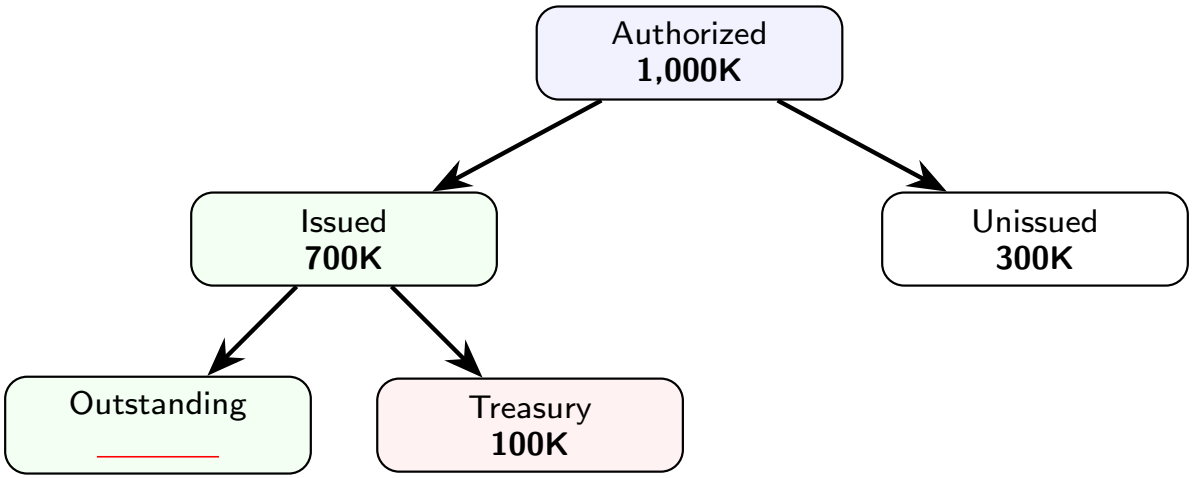
EX. A corporation has authorized 1M shares, issued 700K shares, and repurchased 100K shares as treasury stock.



Shares - Outstanding

- **Outstanding shares**: Shares that are currently held by investors.

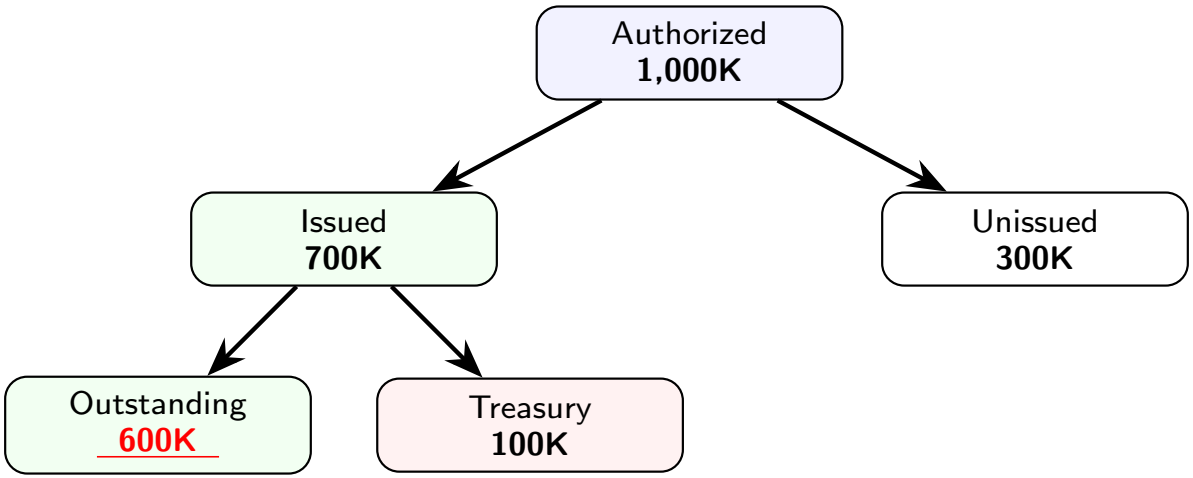
EX. The same corporation has issued 700K shares and holds 100K shares as treasury stock. How many shares are **outstanding**?



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Shares - Comprehensive Example

EX. The balance sheet for Crutcher Corporation reported 200K shares outstanding, 300K shares authorized, and 20K shares in treasury stock. Compute the **maximum number of new shares** that Crutcher could issue.

Type of shares	**Number of shares**
Authorized shares	300,000
Outstanding shares	200,000
Treasury shares	20,000
Issued shares (Outstanding + Treasury)	
Shares available for issuance (Authorized - Issued)	

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Type of shares	**Number of shares**
Authorized shares	300,000
Outstanding shares	200,000
Treasury shares	20,000
Issued shares (Outstanding + Treasury)	220,000
Shares available for issuance (Authorized - Issued)	

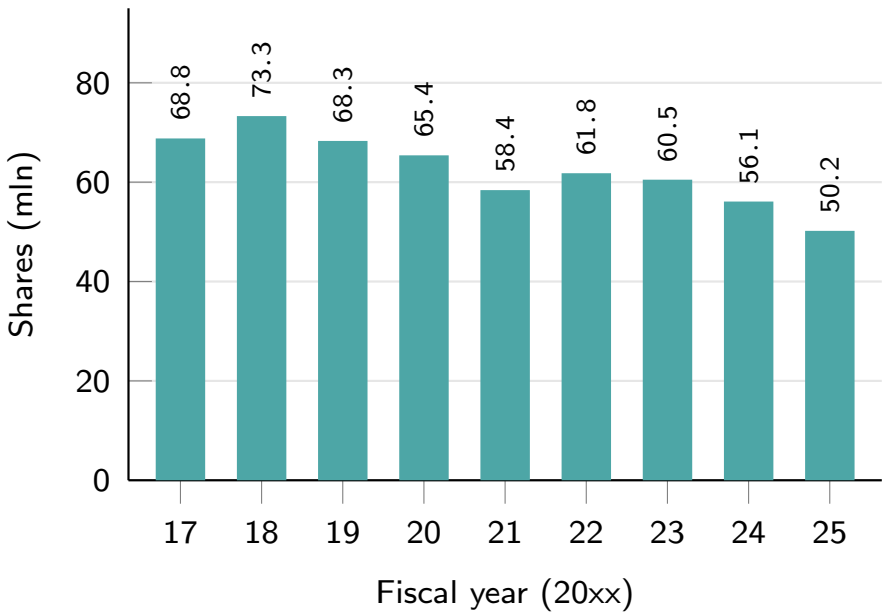
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Outstanding shares	200,000
Treasury shares	20,000
Issued shares (Outstanding + Treasury)	220,000
Shares available for issuance (Authorized - Issued)	80,000

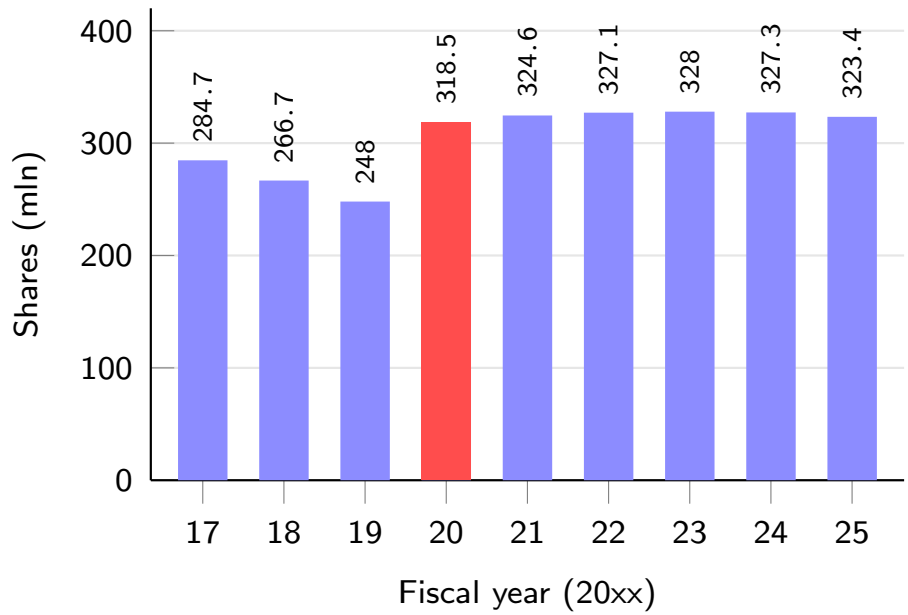
Shares Outstanding in Practice

Crocs (CROX)



Repurchases (treasury stock) ⇒ fewer shares outstanding.

United Airlines (UAL)



A large **issuance** in 2020 (COVID-19) ⇒ more shares outstanding.

Source: each company's 10-K (SEC EDGAR), shares of common stock outstanding reported on the cover page, as of early the following year.

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Dividends

- A distribution of a portion of a corporation's _____ to its shareholders.
- Firms usually pay dividends when they have **excess cash** and **stable earnings**.
- _____.
- Cash payment, though _____ an Expense![4]
- Dividends reduce _____.

Retained Earnings Equation

End. RE = Beg. RE + Net Income - Dividends

[4] Recall: Expenses are those that are incurred in the process of generating revenue. Does Dividends fit this definition?

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- Dividends are paid **only to shareholders**; _____ for treasury stocks (held by the corporation itself).

EX. Netpass Company has 300K shares of common stock authorized, 270K shares issued, and 100K shares of treasury stock. The board of directors declares a dividend of \$1 per share of common stock. **How much is the total dividend?**

Item	Amount
Issued shares	270,000
Treasury shares	(100,000)
Outstanding shares	
Dividend per share	\$1
Total dividends	

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Treasury shares	(100,000)
Outstanding shares	170,000
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The residual interest in the assets of the corporation after deducting liabilities. It represents the ownership interest of shareholders in the corporation.

[5] covered in ACC301 and ACC302.

[6] C.C − T.S matches the **outstanding shares**, those held by shareholders. Treasury shares are held by the company itself, not by shareholders, so they are subtracted from SHE.

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The residual interest in the assets of the corporation after deducting liabilities. It represents the ownership interest of shareholders in the corporation.

- **Contributed capital (C.C)**
- **Treasury stock (T.S)** (a.k.a. stock repurchase; stock buyback)
- **Retained earnings (R.E)**
- Accumulated other comprehensive income[5]

SHE= C.C - T.S + R.E[6]

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Shareholders' Equity: Example

EX. Year 1: a corporation issues **100 shares at $200** per share. Year 2: it repurchases **20 shares at $200** per share (treasury stock). Retained earnings at the end of Year 2: $5,000.

Contributed capital (C.C)	_____
Less: Treasury stock (T.S)	_____
Retained earnings (R.E)	$5,000
SHE	_____

- C.C − T.S = _____, the capital behind the 80 **outstanding shares**.
- What if the repurchase price were above or below $200? Not covered in this course.[7]

[7] Treasury stock is recorded at the repurchase price (cost). The difference from the original issue price is **never a gain or loss** on the income statement, because the company is trading with its own owners; it adjusts contributed capital or retained earnings instead (ACC301/302).

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SHE	**$21,000**

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Contributed Capital

The amount of capital that shareholders have contributed to the corporation.

- In this course, all cash received from issuing shares is recorded in **one account**: **Common Stock** (= contributed capital).[8,9]

EX. A corporation issues 1K shares of common stock at $10 per share.

[8] In practice, accountants split it into **par value** and **additional paid-in capital (APIC)**. The split matters for accounting records only; see the Appendix (not for exam).

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Cash (+A)	10K	
Common Stock (+SH)		10K

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Treasury Stock

Treasury Stock (Contra Equity)

The cost value of shares repurchased by the corporation.

- Shares **repurchased** by the firm; **held** by the firm.
- Share repurchases
  - ⇒ _____.
  - ⇒ Net assets (i.e., SHE = A - L) _____.[10]
  - ⇒ **Treasury shares reduce SHE!**
- Firms are more likely to buy back when the stock price is _____.

[10] Understand it this way: the company is being well managed, so it can do the same with fewer resources (i.e., lower cash); the company can repurchase its own stocks (i.e., pay cash to some investors by buying their shares back). So lower cash; lower assets; lower SHE!

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- Firms are more likely to buy back when the stock price is **low (undervalued)** .

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- Net income (NI): Increases retained earnings.

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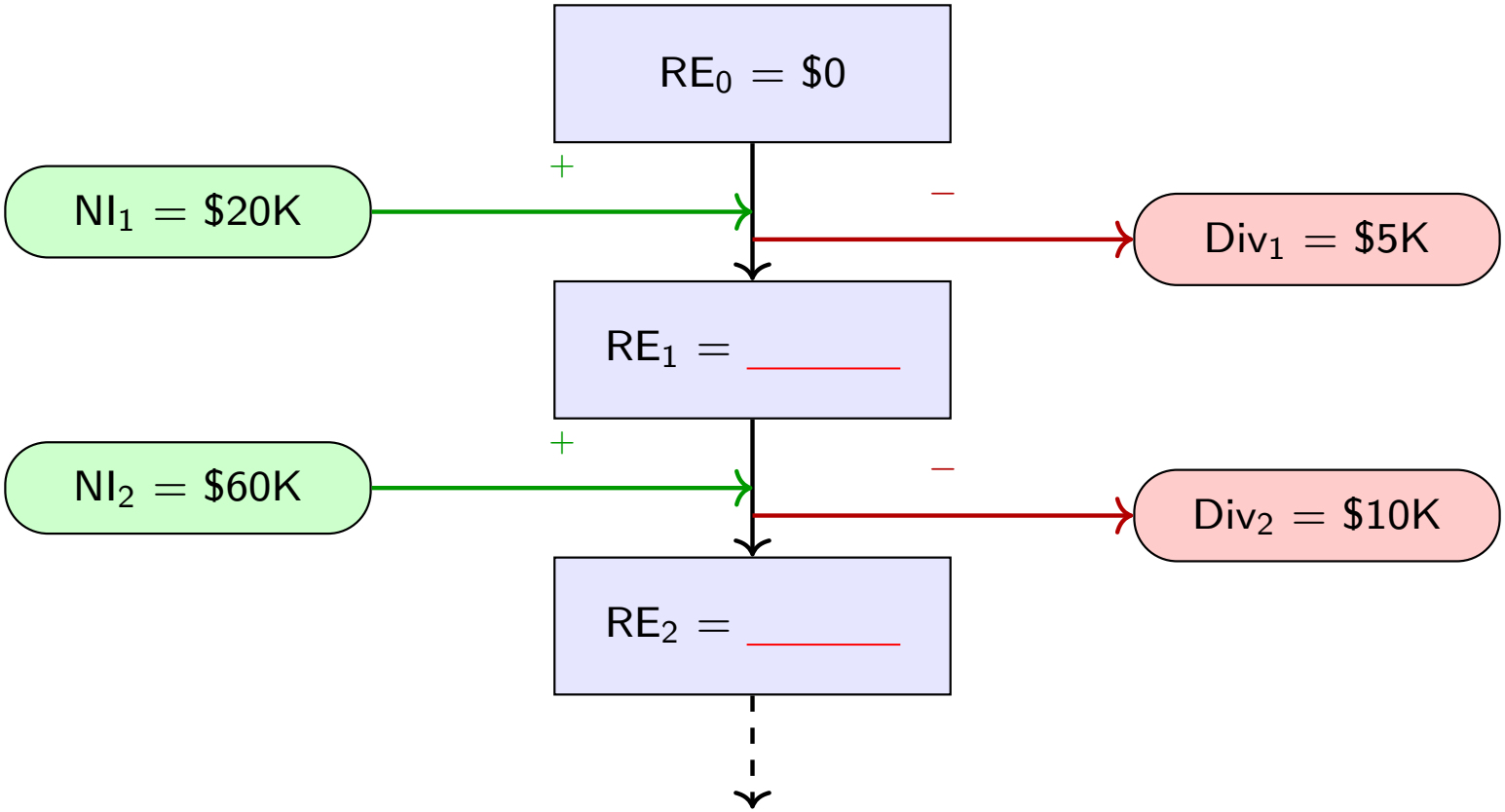
- Retained earnings (RE) equation:

$$\text{Ending RE} = \text{Beginning RE} + \text{NI} - \text{Div}$$

- Net income (NI): Increases retained earnings.
- Dividends (Div): Decrease retained earnings.

Recall: NI, Div, and RE (Ch0)

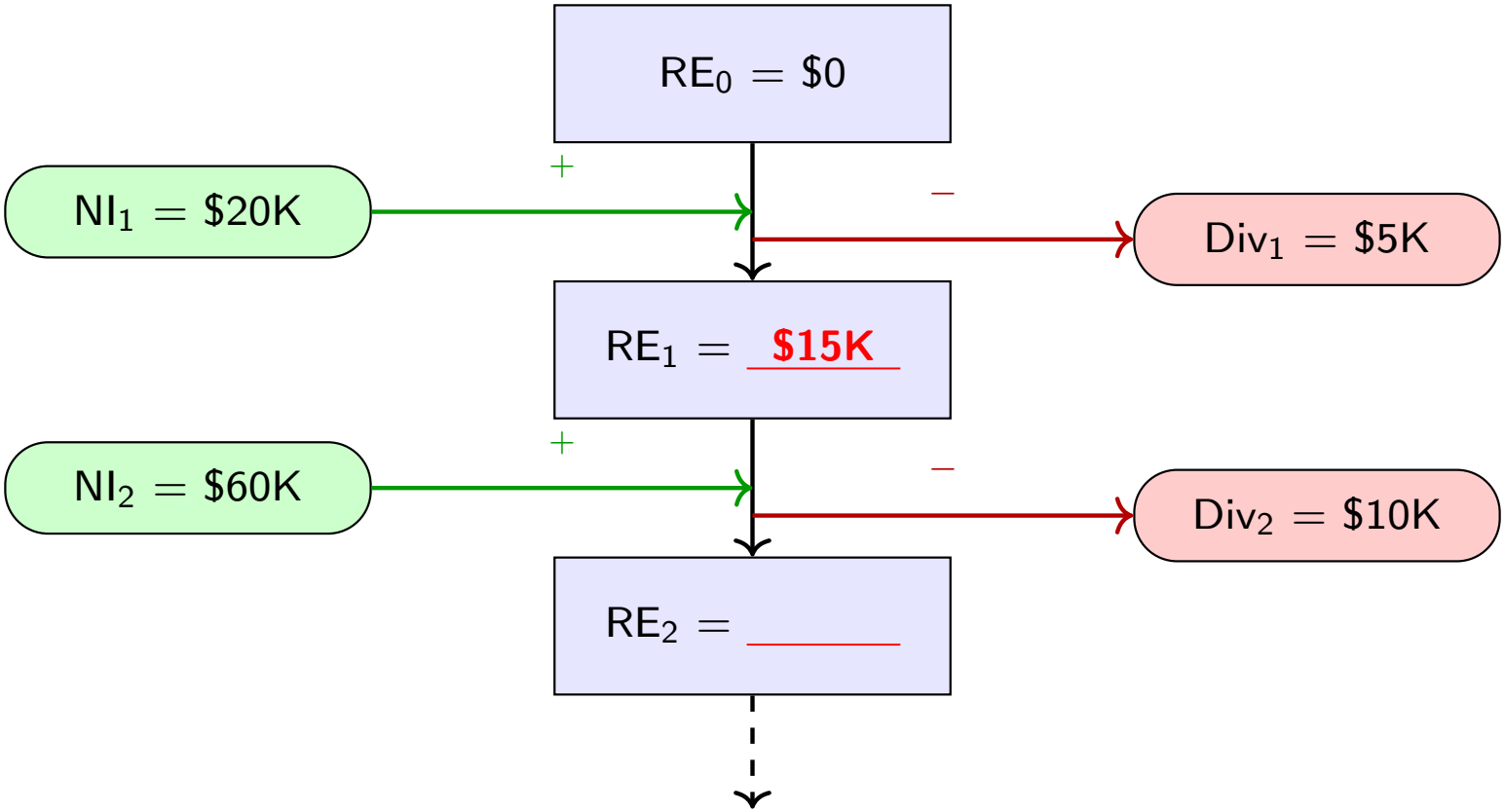
EX. A business opens with $RE_0 = \$0$. Year 1: NI \$20K, Div \$5K. Year 2: NI \$60K, Div \$10K.



Ending RE = Beginning RE + NI − Div

Recall: NI, Div, and RE (Ch0)

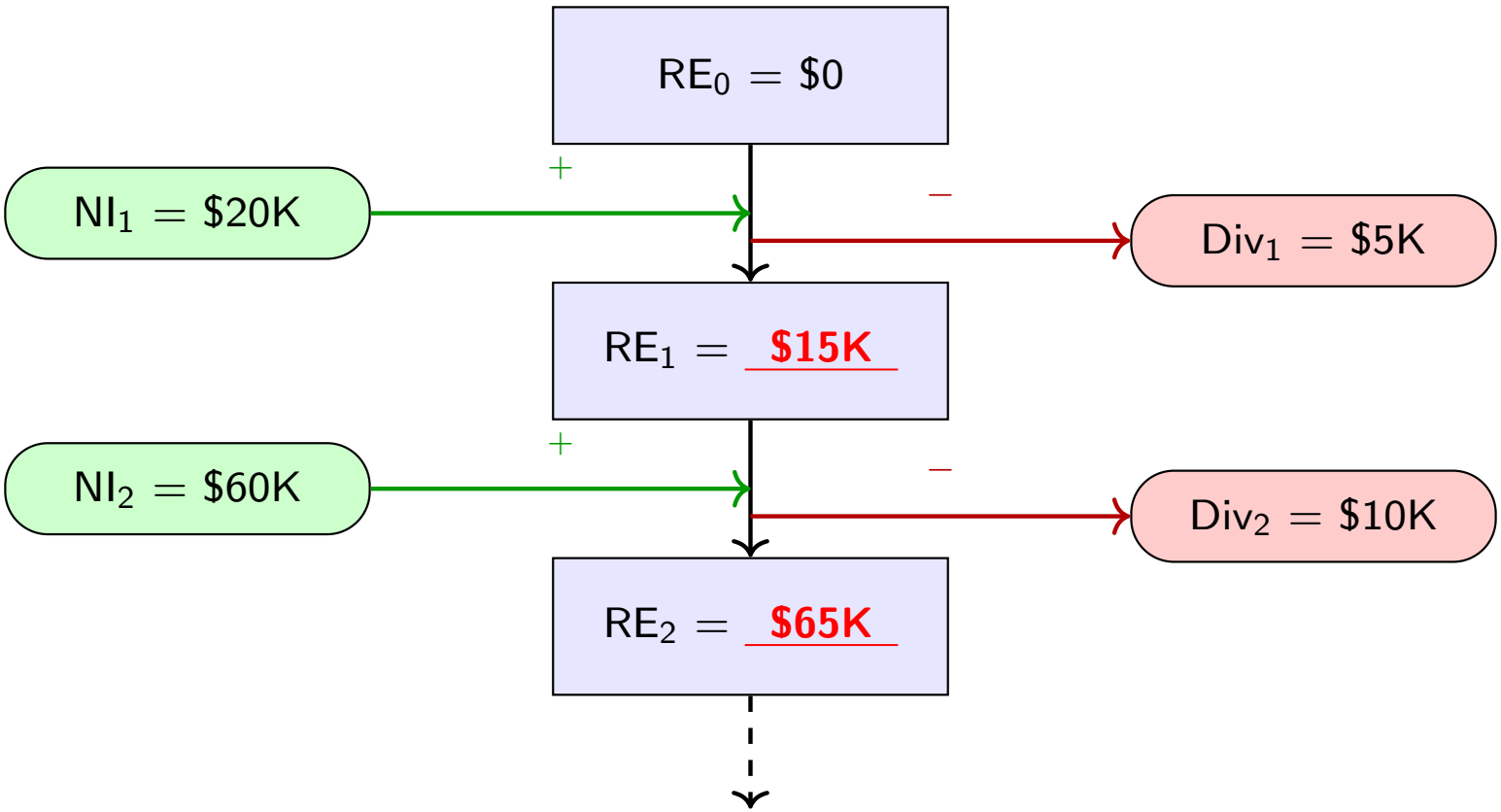
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Ending RE = Beginning RE + NI − Div

Recall: NI, Div, and RE (Ch0)

EX. A business opens with $RE_0 = \$0$. Year 1: NI \$20K, Div \$5K. Year 2: NI \$60K, Div \$10K.



Ending RE = Beginning RE + NI − Div

EX. A corporation has beginning retained earnings of $50,000, net income of $20,000, and dividends of $5K.

Item	Amount ($)
Beginning Retained Earnings	
Add: Net Income	
Less: Dividends	
Ending Retained Earnings	

Table: Statement of Retained Earnings

EX. A corporation has beginning retained earnings of $50,000, net income of $20,000, and dividends of $5K.

Item	Amount ($)
Beginning Retained Earnings	50,000
Add: Net Income	
Less: Dividends	
Ending Retained Earnings	

Table: Statement of Retained Earnings

EX. A corporation has beginning retained earnings of $50,000, net income of $20,000, and dividends of $5K.

Item	Amount ($)
Beginning Retained Earnings	50,000
Add: Net Income	20,000
Less: Dividends	
Ending Retained Earnings	

Table: Statement of Retained Earnings

EX. A corporation has beginning retained earnings of $50,000, net income of $20,000, and dividends of $5K.

Item	Amount ($)
Beginning Retained Earnings	50,000
Add: Net Income	20,000
Less: Dividends	(5,000)
Ending Retained Earnings	

Table: Statement of Retained Earnings

EX. A corporation has beginning retained earnings of $50,000, net income of $20,000, and dividends of $5K.

Item	Amount ($)
Beginning Retained Earnings	50,000
Add: Net Income	20,000
Less: Dividends	(5,000)
Ending Retained Earnings	65,000

Table: Statement of Retained Earnings

1 Corporations

2 Shares

3 Dividends

4 Shareholders' Equity
- Contributed Capital
- Treasury Stock
- Retained Earnings

5 Ratio analysis
- Return on Equity (ROE)
- Earnings Per Share (EPS)
- Price-to-Earnings (PER) Ratio

6 Concluding illustration

7 Appendix: Par Value and APIC (Not for exam)

Return on Equity (ROE)

Return on Equity (ROE)

Return (net income) on \$1 of equity investment.
Calculated as: $\frac{\text{Net Income}}{\text{Average Shareholders' Equity}}$ if Preferred Dividends$= 0$.

- Measures how much, for \$1 of equity investment, the firm generates in net income.
- Higher ROE values generally indicate better performance.

EX. A corporation has net income of \$100K and average shareholders' equity of \$500K.

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EX. A corporation has net income of \$100K and average shareholders' equity of \$500K.

- ROE = \$100K / \$500K = 0.20 or 20%
- This means the corporation generated a return of 20% on its equity investment.

Earnings Per Shares (EPS)

Earnings Per Share (EPS)

A measure of a company's profitability on a per-share basis.
Calculated as: $\frac{\text{Net Income}}{\text{Average Shares Outstanding}}$ if Preferred Dividends$= 0$.

- Measures how much, for each share of common stock, the corporation generates in net income.
- Higher EPS values generally indicate better performance.

EX. A corporation has net income of \$100K and average shares outstanding of 50K.

Earnings Per Shares (EPS)

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A measure of a company's profitability on a per-share basis.
Calculated as: $\frac{\text{Net Income}}{\text{Average Shares Outstanding}}$ if Preferred Dividends$= 0$.

- Measures how much, for each share of common stock, the corporation generates in net income.
- Higher EPS values generally indicate better performance.

EX. A corporation has net income of \$100K and average shares outstanding of 50K.

- EPS $=$ \$100K / 50K $=$ \$2.00
- This means the corporation generated \$2.00 in earnings for each share of common stock outstanding.

Treasury Stock, ROE, and EPS

- Treasury Stock $\Rightarrow$ _____ Shares Outstanding $\Rightarrow$ _____ EPS
- Treasury Stock $\Rightarrow$ _____ SHE $\Rightarrow$ _____ ROE

EX. A firm has net income of \$100K and average shares outstanding of 50K.

- How much is the EPS without stock repurchases
- How much is the EPS if the corporation repurchases 10K shares as treasury stock?

Treasury Stock, ROE, and EPS

- Treasury Stock ⇒ **Lower** Shares Outstanding ⇒ **Higher** EPS
- Treasury Stock ⇒ _____ SHE ⇒ _____ ROE

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  - EPS = $100K / 50K = $2.00
- How much is the EPS if the corporation repurchases 10K shares as treasury stock?
  - EPS = $100K / (50K - 10K) = $2.50

Treasury Stock, ROE, and EPS

- Treasury Stock ⇒ **Lower** Shares Outstanding ⇒ **Higher** EPS
- Treasury Stock ⇒ **Lower** SHE ⇒ **Higher** ROE

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- How much is the EPS without stock repurchases
  - EPS = $100K / 50K = $2.00
- How much is the EPS if the corporation repurchases 10K shares as treasury stock?
  - EPS = $100K / (50K - 10K) = $2.50

⇒ Purchasing its own stock can **boost the firm's financial performance metrics**!

Price-to-Earnings (P/E) Ratio

Which one is cheaper? (assuming that they have the same growth potential)

Q1. Same EPS with different Price

- A firm that generates \$10 EPS; its stock price is now \$100.
- A firm that generates \$10 EPS; its stock price is now \$50. → _____

Q2. Same Price with different EPS

- A firm that generates \$10 EPS; its stock price is now \$100.
- A firm that generates \$20 EPS; its stock price is now \$100. → _____

Q3. Different EPS and Price

- A firm that generates \$10 EPS; its stock price is now \$100.
- A firm that generates \$5 EPS; its stock price is now \$20. → _____

⇒ The **P/E ratio** can answer this!

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Which one is cheaper? (assuming that they have the same growth potential)

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- A firm that generates \$10 EPS; its stock price is now \$50. → **cheaper**

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- A firm that generates \$10 EPS; its stock price is now \$100.
- A firm that generates \$20 EPS; its stock price is now \$100. → **cheaper**

Q3. Different EPS and Price

- A firm that generates \$10 EPS; its stock price is now \$100.
- A firm that generates \$5 EPS; its stock price is now \$20. → **cheaper**

⇒ The **P/E ratio** can answer this!

Price-to-Earnings (P/E) Ratio

Price-to-Earnings (P/E) Ratio

A valuation ratio that compares a company's current share price to its EPS.
Calculated as: $\frac{\text{Stock Price per Share}}{\text{Earnings Per Share (EPS)}}$

- Measures how much investors are willing to pay for each dollar of earnings.
- Higher P/E ratios indicate currently higher prices relative to earnings, i.e., Premium!
- Why?
  1. the stock is overvalued.
  2. investors expect that future earnings will be higher.

Comparison of United Airlines and Delta Airlines (**End of Nov 2025**)

Metric	**UAL**	**DAL**
Stock Price	\$92.20	\$58.57
2025 EPS forecast from CNBC	\$10.82	\$6.02
P/E	8.52	9.73

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Metric	**UAL**	**DAL**
Stock Price	\$92.20	\$58.57
2025 EPS forecast from CNBC	\$10.82	\$6.02
P/E	8.52	9.73

- Assume they have similar growth prospects. Which one is undervalued?

Comparison of United Airlines and Delta Airlines (**End of Nov 2025**)

Metric	**UAL**	**DAL**
Stock Price	\$92.20	\$58.57
2025 EPS forecast from CNBC	\$10.82	\$6.02
P/E	8.52	9.73

- Assume they have similar growth prospects. Which one is undervalued? United Airlines (with lower P/E).

1 Corporations

2 Shares

3 Dividends

4 Shareholders' Equity
- Contributed Capital
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7 Appendix: Par Value and APIC (Not for exam)

B/S example 1: Growing Retained earnings

- Year 1: Raised $1M in capital
- Year 1: Borrowed $2M from a bank
- Year 1: Earned $200K in net income.
- Year 2: Earned $300K in net income.
- Year 3: Earned $400K in net income.
- **Calculate Retained Earnings (RE) at the end of each year.**[11]

Balance Sheet	Year 1	Year 2	Year 3
Assets ...	...	...	...
Total Assets			
Liabilities and Shareholders' Equity Liabilities ...	...	...	...
Total Liabilities	2,000,000	2,000,000	2,000,000
Shareholders' Equity Contributed Capital (C.C) Retained Earnings (R.E)	1,000,000	1,000,000	1,000,000
Total Shareholders' Equity			
Total Liabilities and Shareholders' Equity			

[11] The below B/S can be made from the information above.

B/S example 1: Growing Retained earnings

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- **Calculate Retained Earnings (RE) at the end of each year.**[11]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	3,200,000		
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	2,000,000
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	1,000,000
Retained Earnings (R.E)	200,000		
Total Shareholders' Equity	1,200,000		
Total Liabilities and Shareholders' Equity	3,200,000		

[11] The below B/S can be made from the information above.

B/S example 1: Growing Retained earnings

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- Year 3: Earned $400K in net income.
- **Calculate Retained Earnings (RE) at the end of each year.**[11]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	3,200,000	3,500,000	
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	2,000,000
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	1,000,000
Retained Earnings (R.E)	200,000	500,000	
Total Shareholders' Equity	1,200,000	1,500,000	
Total Liabilities and Shareholders' Equity	3,200,000	3,500,000	

[11] The below B/S can be made from the information above.

B/S example 1: Growing Retained earnings

- Year 1: Raised $1M in capital
- Year 1: Borrowed $2M from a bank
- Year 1: Earned $200K in net income.
- Year 2: Earned $300K in net income.
- Year 3: Earned $400K in net income.
- **Calculate Retained Earnings (RE) at the end of each year.**[11]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	3,200,000	3,500,000	3,900,000
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	2,000,000
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	1,000,000
Retained Earnings (R.E)	200,000	500,000	900,000
Total Shareholders' Equity	1,200,000	1,500,000	1,900,000
Total Liabilities and Shareholders' Equity	3,200,000	3,500,000	3,900,000

[11] The below B/S can be made from the information above.

B/S example 2: Growing Retained Earnings even with dividends

- Year 1: Raised $1M in capital
- Year 1: Borrowed $2M from a bank
- Year 1: Earned $200K in net income.
- Year 2: Earned $300K in net income.
- Year 3: Earned $400K in net income.
- Year 3: **Paid $250K dividends**.
- **Calculate Retained Earnings (RE) at the end of each year.**[12]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	3,200,000	3,500,000	
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	
Retained Earnings (R.E)	200,000	500,000	
Total Shareholders' Equity	1,200,000	1,500,000	
Total Liabilities and Shareholders' Equity	3,200,000	3,500,000	

[12] The below B/S can be made from the information above.

B/S example 2: Growing Retained Earnings even with dividends

- Year 1: Raised $1M in capital
- Year 1: Borrowed $2M from a bank
- Year 1: Earned $200K in net income.
- Year 2: Earned $300K in net income.
- Year 3: Earned $400K in net income.
- Year 3: **Paid $250K dividends**.
- **Calculate Retained Earnings (RE) at the end of each year.**[12]

Balance Sheet	Year 1	Year 2	Year 3
Assets			
...	...	...	...
Total Assets	3,200,000	3,500,000	3,650,000
Liabilities and Shareholders' Equity			
Liabilities			
...	...	...	...
Total Liabilities	2,000,000	2,000,000	2,000,000
Shareholders' Equity			
Contributed Capital (C.C)	1,000,000	1,000,000	1,000,000
Retained Earnings (R.E)	200,000	500,000	650,000
Total Shareholders' Equity	1,200,000	1,500,000	1,650,000
Total Liabilities and Shareholders' Equity	3,200,000	3,500,000	3,650,000

[12] The below B/S can be made from the information above.

B/S example 3: Growing Retained Earnings even with dividends

- Year 1: Raised $1M in capital.
- Year 1: Borrowed $2M from a bank.
- Year 1: Reported **$600K losses**.
- Year 2: Reported **$800K losses**.
- **Calculate Retained Earnings (RE) at the end of each year.**[13]

Balance Sheet	Year 1	Year 2
Assets		
...	...	...
Total Assets	2,400,000	
Liabilities and Shareholders' Equity		
Liabilities		
...	...	...
Total Liabilities	2,000,000	
Shareholders' Equity		
Contributed Capital (C.C)	1,000,000	
Retained Earnings (R.E)	(600,000)	
Total Shareholders' Equity	400,000	
Total Liabilities and Shareholders' Equity	2,400,000	

- If Retained Earnings is negative, it is called **accumulated deficit**.
- The company has an **accumulated deficit** of **(400K)**.

[13] The below B/S can be made from the information above.

B/S example 3: Growing Retained Earnings even with dividends

- Year 1: Raised $1M in capital.
- Year 1: Borrowed $2M from a bank.
- Year 1: Reported **$600K losses**.
- Year 2: Reported **$800K losses**.
- **Calculate Retained Earnings (RE) at the end of each year.**[13]

Balance Sheet	Year 1	Year 2
Assets		
...	...	...
Total Assets	2,400,000	1,600,000
Liabilities and Shareholders' Equity		
Liabilities		
...	...	...
Total Liabilities	2,000,000	2,000,000
Shareholders' Equity		
Contributed Capital (C.C)	1,000,000	1,000,000
Retained Earnings (R.E)	(600,000)	(1,400,000)
Total Shareholders' Equity	400,000	(400,000)
Total Liabilities and Shareholders' Equity	2,400,000	1,600,000

- If Retained Earnings is negative, it is called **accumulated deficit**.
- The company has an **accumulated deficit** of **(400K)**.

[13] The below B/S can be made from the information above.

Appendix (Not for exam): Table of Contents

1. Corporations
2. Shares
3. Dividends
4. Shareholders' Equity
   - Contributed Capital
   - Treasury Stock
   - Retained Earnings
5. Ratio analysis
   - Return on Equity (ROE)
   - Earnings Per Share (EPS)
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Appendix (Not for exam): Par Value and APIC

Common Stock, Par Value (Equity)

The nominal value assigned to each share of stock.

- It used to be a legal requirement
- But now, it is often set at a very low amount (e.g., \$0.01).
- This is not the **market value** of the stock.

Additional Paid-in Capital (APIC, Equity)

The amount received from shareholders in excess of the par value of the stock.

EX. Issue 1K shares, par \$1, at \$10 per share.

Cash (+A)	10K
Common Stock (+SH)	1K
APIC (+SH)	9K

EX. Issue 2K shares, par \$2, at \$15 per share.

Cash (+A)	30K
Common Stock (+SH)	4K
APIC (+SH)	26K

Common Stock + APIC = the **same total** recorded as Common Stock in this course (\$10K and \$30K).